

Paper Practice 1.1 • WORKED KEY**Numerical Expressions**

WORKED KEY (odd-numbered items) – Show all of your work in the space provided. No calculator. For translation problems, write the expression FIRST, then evaluate it. Circle your final answers.

- 1.** Simplify each expression. (Warm-Up Review)

a. $-3 + (-14)$

Answer: -17

b. $15 - 22 + (-6)$

Answer: $15 - 22 = -7$; $-7 + (-6) = -13$

c. $-9 - (-5) - (-3)$

Answer: $-9 + 5 + 3 = -1$

- 2.** Write a numerical expression for each phrase.

Do NOT evaluate. (Translate)

a. four plus twelve divided by six

b. six times eleven minus three times seven

c. fifteen less than three times eight

- 3.** Write a numerical expression for each phrase, using grouping symbols where they are needed. Then evaluate. (Translate)

a. the sum of seven and eleven, divided by three

Answer: $(7 + 11) \div 3 = 18 \div 3 = 6$

b. five times the difference of twelve and four

Answer: $5(12 - 4) = 5(8) = 40$

c. twice the sum of nine and six

Answer: $2(9 + 6) = 2(15) = 30$

- 4.** Write and evaluate a numerical expression for EACH phrase below, then explain in one sentence why the two answers are different. (Compare)

a. the sum of eight and four, divided by two

b. eight plus four divided by two

5. A bowling handicap is found by subtracting the player's average from 220, multiplying that result by 3, and then dividing by 4. Write a numerical expression for the handicap of a player whose average is 180, then evaluate it. (Write & Evaluate)

Answer: $3(220 - 180) \div 4 = 3(40) \div 4 = 120 \div 4 = 30$

6. Answer each part. (Powers)

- Write $5 \cdot 5 \cdot 5 \cdot 5$ using an exponent, then name the base and the exponent.
- Evaluate 3^4
- Evaluate 2^6

10. Evaluate: $(3^3 - 7) / (12 + 8)$ (GEMS)

11. Error Analysis – Gus evaluated the expression below and wrote 16. Identify his mistake, then evaluate correctly. (Error Analysis)

Gus's work: $4^2 - 6 \div 2 + 3 = 16 - 3 + 3 = 16$

Answer: He ignored the fraction bar, which is a GROUPING symbol – it groups the entire numerator and the entire denominator. Correct: $(4^2 - 6) \div (2 + 3) = (16 - 6) \div 5 = 10 \div 5 = 2$

7. Evaluate: $30 - 4 + 5^2 - 6 \cdot 9$ (GEMS)

Answer: exponent first: $30 - 4 + 25 - 54$; then multiply: $6 \cdot 9 = 54$; left to right: $26 + 25 = 51$, $51 - 54 = -3$

8. Evaluate: $(8 + 4)^2 / 4(9 - 3)$ (GEMS)

9. Evaluate: $20 - [8 + (5 - 3)^2] + 7$ (GEMS)

Answer: innermost first: $(5 - 3)^2 = 2^2 = 4$; $[8 + 4] = 12$; $20 - 12 + 7 = 15$

12. Mr. Nappier's class is planning a field trip for 24 students. The costs are shown in the table. Write a numerical expression for the total cost of the bus, admission, and lunch for every student, then evaluate it. If the class budget is \$650, can they afford it? Explain. (Write & Evaluate)

Item	Cost
Bus (flat fee)	\$180
Admission	\$12 per student
Lunch	\$8 per student
T-shirt	\$6 per student